



Computational Fluid Dynamics; Abdominal Aortic Aneurysm Observational Model

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Abstract

This report presents a Computational Fluid Dynamics (CFD) investigation of blood flow within an idealised Abdominal Aortic Aneurysm (AAA) using ANSYS Fluent 2025 R2. Steady uniform, steady parabolic and transient periodic inlet conditions were applied to examine the influence of inlet boundary conditions on aneurysmal haemodynamics. Blood was modelled as an incompressible Newtonian fluid under laminar flow conditions and post-

processing was performed using both Fluent and ANSYS CFD-Post. The simulations demonstrated the formation of a high velocity central jet surrounded by regions of low-velocity recirculating flow within the aneurysm sac. The steady parabolic inlet profile produced stronger near wall velocity gradients and more pronounced recirculation structures compared with the uniform inlet condition. The transient periodic simulation generated continuously evolving vortex structures and

time-dependent wall shear stress distributions throughout the cardiac cycle. Regions of low and reversed wall shear stress were observed within the aneurysm bulge due to flow separation and near wall recirculation. These disturbed haemodynamic conditions are significant because they are associated with thrombus formation, endothelial dysfunction and aneurysm progression in abdominal aortic aneurysms [1–4]. Overall, the transient simulation provided the most physiologically realistic representation of flow behaviour within the aneurysm geometry.

Introduction

The lab simulated blood flow through a simple digital model of an Abdominal Aortic Aneurysm (AAAs). These are localized dilations of the abdominal aorta that significantly alter normal blood flow behaviour. The disturbed haemodynamic within aneurysmal regions can produce flow separation, recirculation and vortex formation, which in turn leads to variations in wall shear stress (WSS). These effects are of clinical importance, as abnormal WSS and regions of stagnant flow have been associated with aneurysm growth, thrombus formation and increased rupture risk [1-3].

The approach taken here was utilising Computational Fluid Dynamics (CFD), which is an effective method for investigating blood flow within vascular geometries by numerically solving the governing fluid flow equations. CFD enables the detailed analysis of velocity distributions, pressure variations and transient flow structures that are more difficult to obtain experimentally. In cardiovascular engineering, CFD is widely used to examine how geometry and inlet boundary conditions influence haemodynamic behaviour [1].

In this study the CFD simulations were performed on an idealised abdominal aortic aneurysm geometry using ANSYS Fluent. Blood was modelled as an incompressible Newtonian fluid under laminar flow conditions, wherein three inlet conditions were investigated: a steady uniform inlet profile, a steady parabolic inlet profile and a transient periodic inlet waveform which represented physiological pulsatile flow. The influence of these boundary conditions on velocity fields, recirculation zones and wall shear stress distributions within the aneurysm was analysed.

Post-processing was carried out using ANSYS CFD-Post in order to best visualise transient flow behaviour throughout the cardiac cycle.

Objectives

The primary objective of this study was to investigate the haemodynamic behaviour of blood flow within an idealised abdominal aortic aneurysm (AAA) by using Computational Fluid Dynamics (CFD) simulations performed in ANSYS Fluent. The aim was to examine how different inlet boundary conditions influenced the resulting velocity distributions, flow structures and wall shear stress patterns within the aneurysm geometry.

Initially, a steady laminar flow simulation with a uniform inlet velocity profile was to be analysed in order to establish the baseline flow behaviour throughout the vessel.

A fully developed parabolic inlet profile was then implemented through a User Defined Function (UDF) to investigate the effect of a more realistic inlet condition on both the developing flow field as well as the recirculation regions within the aneurysm.

Finally, a transient periodic inlet waveform representing physiological pulsatile blood

flow was applied using a Fourier series based UDF to model the time dependent haemodynamic behaviour which occurs throughout the cardiac cycle.

Velocity contours, vector plots and wall shear stress distributions using both Fluent and ANSYS CFD-Post were to be formed in order to analyse and identify any recirculation zones, regions of flow separation, transient vortex structures and variations in wall shear stress throughout the aneurysm. The simulations were also used to examine the influence of aneurysm geometry on the internal flow behaviour under both steady and transient flow conditions.

Methodology

All of the CFD simulations were performed using ANSYS Fluent 2025 R2 on an idealised, three dimensional abdominal aortic aneurysm (AAA) geometry which was supplied as a pre-generated mesh file. The geometry consisted of a symmetric aneurysm with a vessel diameter of 20 mm at the inlet and outlet and an enlarged aneurysm region with a maximum diameter of approximately 60 mm. The computational mesh consisted primarily of structured hexahedral elements, containing approximately 166,000 cells. The flow direction was aligned with the positive x – axis, with the coordinate origin positioned at the centre of the inlet face.

Blood was modelled as an incompressible Newtonian fluid with a density of 1050 kg/m^3 and a dynamic viscosity of $0.0035 \text{ Pa} \cdot \text{s}$. A pressure-based laminar solver was used throughout all simulations, as the Reynolds number within the vessel was expected to remain below the turbulent transition range. Gravity effects were neglected.

Steady Uniform Inlet Simulation

The initial simulation was performed as a steady-state analysis using a uniform inlet velocity profile. A velocity inlet boundary condition of 0.0565 m/s was applied uniformly across the inlet surface, corresponding to the mean velocity obtained from the Scotti waveform data. A pressure outlet boundary condition with a gauge pressure of 0 Pa was specified at the outlet, while a stationary no-slip wall condition was applied to the aneurysm wall.

The SIMPLE pressure-velocity coupling scheme was selected together with second-order spatial discretisation schemes for both pressure and momentum equations. Prior to solution, the flow field was initialised using the standard initialisation method with values computed from the inlet boundary condition. The inlet velocity was therefore used as the initial estimate throughout the computational domain.

Solution convergence was monitored using scaled residual plots for continuity and the x , y and z -momentum equations. Residual convergence criteria of 1×10^{-3} were specified for all governing equations. The calculation was run for up to 1000 iterations, until all residuals fell below the specified threshold. Velocity monitor points and report definitions were also used to assess the stability of the solution throughout the calculation.

Post-processing for Part 3.1 was initially performed within Fluent. Plane surfaces were generated through the aneurysm geometry in the x , y and z -directions in order to visualise the internal flow field. Velocity contour plots were used to examine the central jet structure and low-velocity regions within the aneurysm bulge. Vector plots were generated to identify recirculation regions and vortical flow structures within the aneurysm. Wall shear stress contours were also examined to

identify regions of elevated and reduced wall loading. In addition, the displayed wall shear stress variable was changed to the x -component of wall shear stress to identify regions of negative shear stress associated with any reversed flow and flow recirculation within the aneurysm.

Steady Parabolic Inlet Simulation

For Part 3.2, the steady-state simulation procedure from Part 3.1 was retained, however the uniform inlet velocity profile was replaced with a fully developed parabolic inlet profile implemented using a User Defined Function (UDF). The UDF was compiled and loaded within Fluent using the built-in compiled UDF functionality. The inlet boundary condition was then modified to use the compiled UDF profile instead of the constant inlet velocity.

Following reinitialization of the flow field from the inlet boundary, the simulation was repeated using the same numerical schemes, material properties and convergence criteria as Part 3.1. Velocity contours and vector plots were again analysed in order to compare the influence of the parabolic inlet profile on the developing flow structures and recirculation behaviour within the aneurysm.

Transient Periodic Flow Simulation

For Part 3.3, the solver type was changed from steady state to transient flow in order to model the physiological pulsatile blood flow through the aneurysm. The inlet boundary condition was modified from the steady parabolic profile to a periodic transient inlet waveform implemented using a Fourier-series-based UDF generated from Scotti flow data. The transient solver employed a second-order implicit formulation. Points were marked at the inlet, outlet, centre of the mesh and at a random point within the bulge for surveying.

The simulation timestep size was set to 0.05 s and multiple cardiac cycles were simulated to achieve periodic flow behaviour. Each transient timestep was iterated for 30 iterations with no convergence tolerance, to ensure there was convergence within each timestep. Velocity monitor points were created at the inlet, outlet, aneurysm centre and within the aneurysm bulge to monitor transient flow behaviour throughout the cardiac cycle. Surface report definitions using vertex-averaged velocity were plotted simultaneously on a shared transient monitor plot to compare the temporal response of velocity at different locations within the geometry.

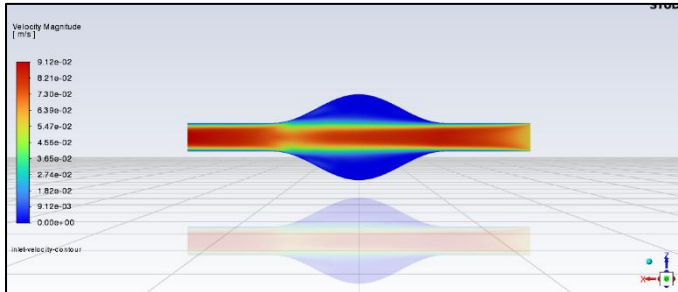
During the final cardiac cycle, transient data files were automatically saved at each timestep using the Fluent autosave functionality. These timestep results were subsequently imported into ANSYS CFD-Post for transient post-processing and animation generation. Velocity vectors, velocity contours and wall shear stress distributions were analysed throughout the cardiac cycle in order to identify transient vortex development, recirculation behaviour and regions of reversed wall shear stress within the aneurysm.

Numerical Results

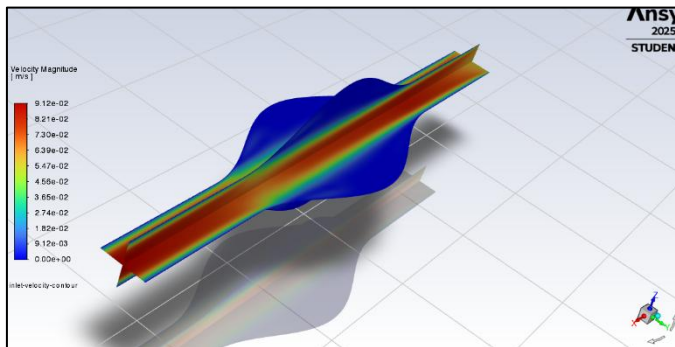
Steady Uniform Inlet Simulation

The initial steady-state simulation using a uniform inlet velocity profile produced a symmetric flow field throughout the aneurysm geometry. Velocity contour plots generated on planes passing through the centre of the aneurysm showed a high-velocity central jet spanning the axial direction of the vessel. As the flow entered the enlarged aneurysm region, the velocity magnitude within the bulge decreased significantly, producing low-velocity regions adjacent to the aneurysm wall. Additionally, the pressure through the mesh dropped significantly on entering the bulge, the

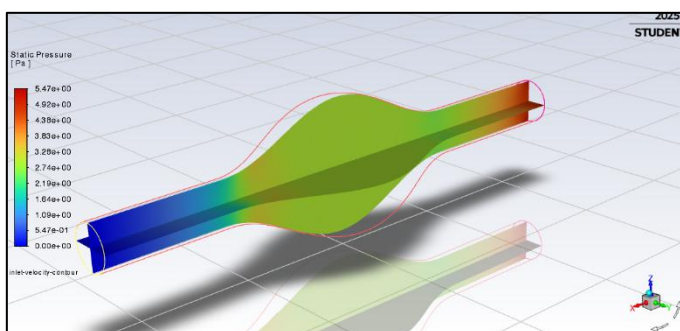
highest pressure occurring near the inlet and the lowest pressure occurring at the outlet where the gauge pressure boundary condition was specified as zero. The flow accelerated again as the vessel diameter reduced towards the outlet.



[Figure 1.1]: Y-normal plane velocity contour with positive x-direction set right to left



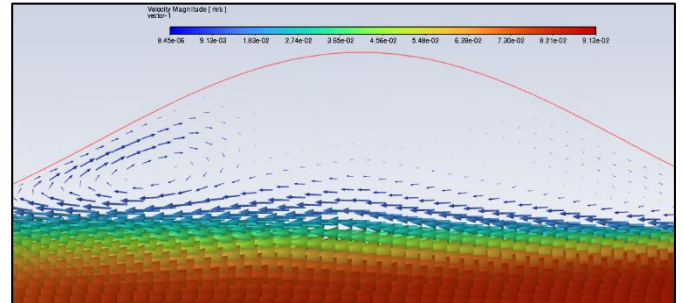
[Figure 1.2]: Angled view of y and z-normal planes being shown with positive x going from back right to left.



[Figure 1.3]: Pressure contour at y and z normal planes.

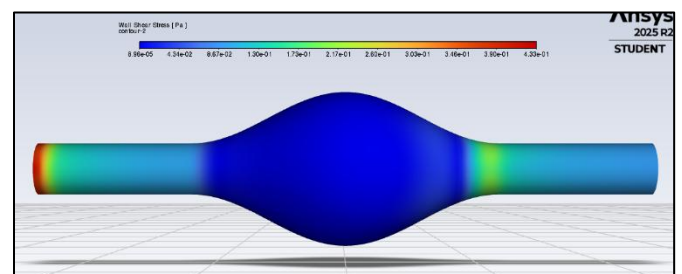
Recirculation zones within the aneurysm bulge were visualised through vector plots of the flow field. While much of the flow concentrated within the central jet, the surrounding fluid near the aneurysm wall exhibited vortical recirculation patterns.

These circulating flow regions were particularly visible within the enlarged aneurysm where local flow separation occurred due to the sudden expansion in vessel diameter.



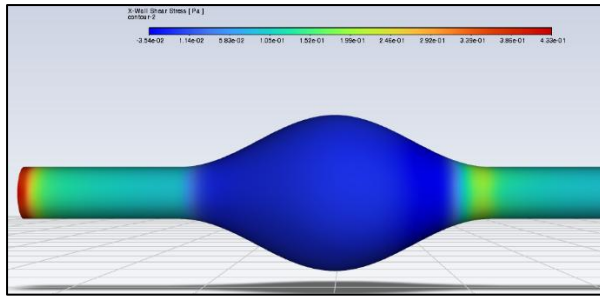
[Figure 1.4]: Velocity vector plot showing recirculation zones within the aneurysm.

Wall shear stress (WSS) contours demonstrated elevated shear stress values within the narrow inlet and outlet sections of the vessel where the velocity gradients near the wall were greatest. In contrast to this, there were significantly lower wall shear stress values observed within the aneurysm bulge due to the reduced near-wall velocities and flow recirculation which was present within this region.



[Figure 1.5]: Wall shear stress contour plot for the steady uniform inlet simulation.

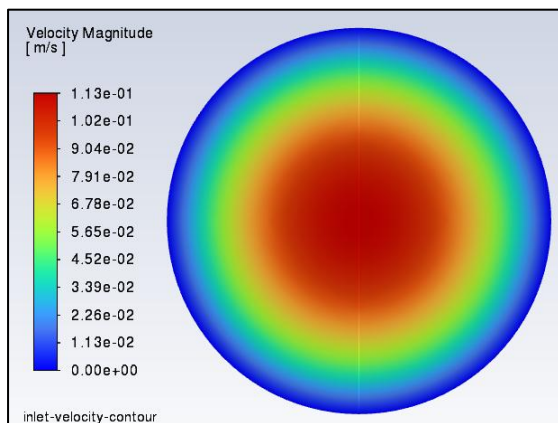
Further analysis of the x-component of wall shear stress revealed regions of negative shear stress within the aneurysm bulge. These negative values indicated local reversal of near-wall flow direction caused by recirculating vortical structures within the aneurysm. The appearance of reversed wall shear stress highlighted the disturbed haemodynamic environment generated within the aneurysm geometry.



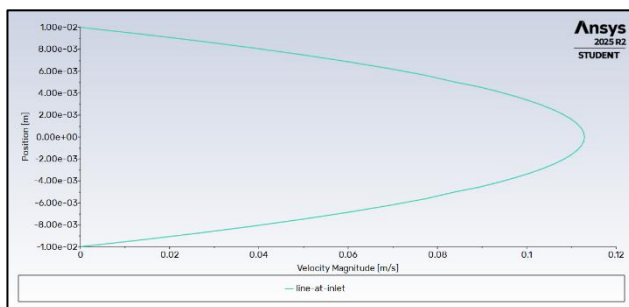
[Figure 1.6]: X-wall shear stress contour plot showcasing negative values.

Steady Parabolic Inlet Simulation

Following implementation of the parabolic inlet velocity profile through the User Defined Function (UDF), the inlet flow field exhibited a fully developed velocity distribution with maximum velocity occurring at the vessel centreline and zero velocity occurring at the wall due to the no-slip condition.



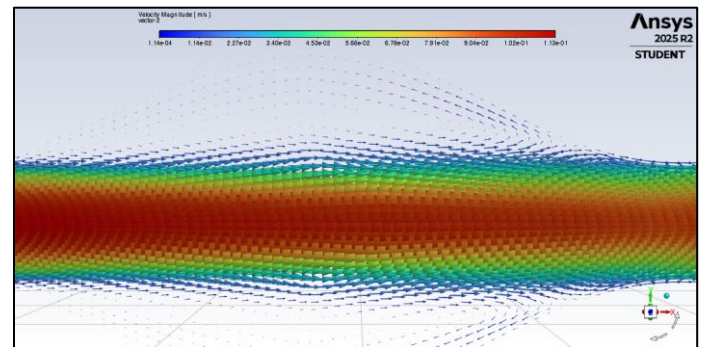
[Figure 2.1]: Parabolic inlet velocity contour.



[Figure 2.2]: Parabolic inlet velocity plot.

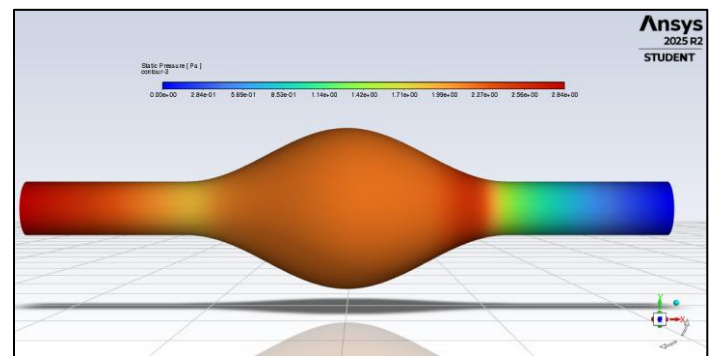
Comparison of the velocity contours and vector plots between the uniform and parabolic inlet conditions showed that the parabolic profile produced a more concentrated centreline jet entering the

aneurysm. This resulted in stronger velocity gradients and slightly more pronounced recirculation structures within the aneurysm bulge. However, the overall flow behaviour remained similar, with the dominant flow structure of a high-velocity central jet surrounded by low-velocity recirculating flow regions.



[Figure 2.3]: Velocity vector plot for Steady Parabolic flow.

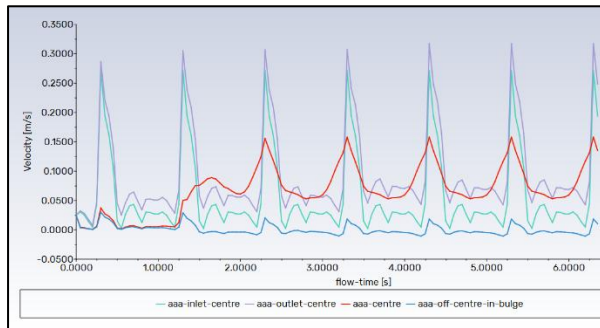
Wall shear stress distributions for the parabolic inlet case also showed similar overall behaviour to the uniform inlet simulation, although the near-wall gradients close to the inlet and bulging regions were definitively more pronounced due to the developed inlet profile.



[Figure 2.1]: Parabolic inlet wall shear stress contour.

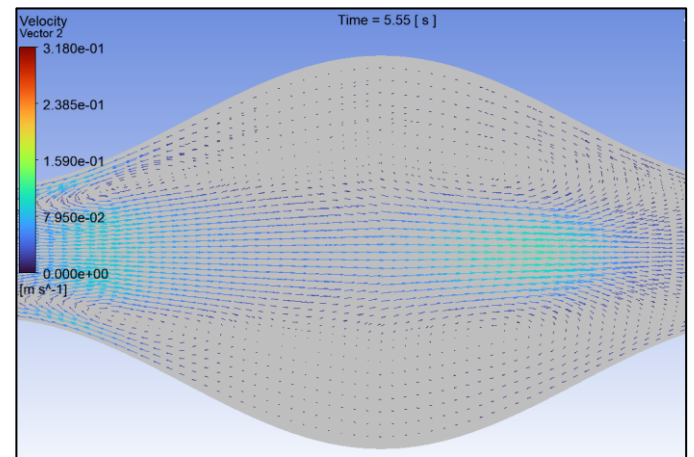
Transient Periodic Flow Simulation

The transient periodic simulation differed from the previous two as it produced time dependent flow structures throughout the cardiac cycle. Velocity monitor plots generated at the inlet, outlet, aneurysm centre and within the aneurysm bulge showed variation in flow velocity as a pulsatile waveform propagated through the geometry. The inlet and outlet velocities responded rapidly to changes in the pulsatile waveform, while the velocity within the aneurysm bulge exhibited delayed and damped behaviour due to the recirculating flow structures present within the enlarged region.

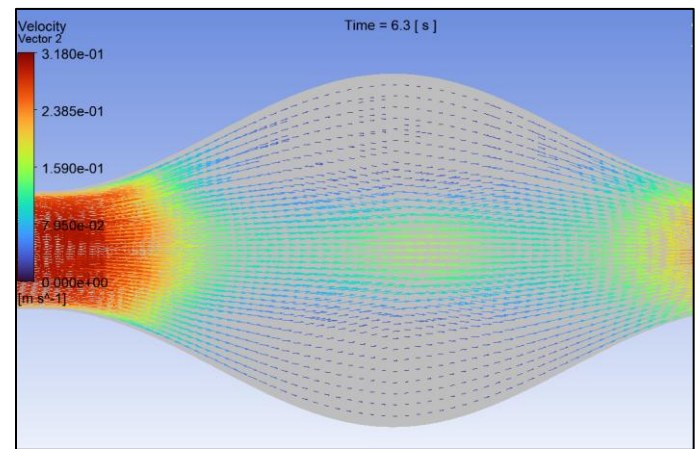


[Figure 2.4]: *Parabolic inlet velocity plot.*

Transient vector plots generated in ANSYS CFD-Post showed the development and downstream movement of vortex structures throughout the cardiac cycle. During high inlet velocity conditions, a strong central jet develops through the aneurysm while slight recirculation regions form near the vessel walls due to flow separation caused by the sudden expansion in geometry. At lower inlet velocities, the central jet weakens and the recirculating flow structures become more dominant throughout the aneurysm sac. These results highlight how the pulsatile flow conditions can produce continuously changing vortex structures and complex haemodynamic behaviour within the aneurysm.

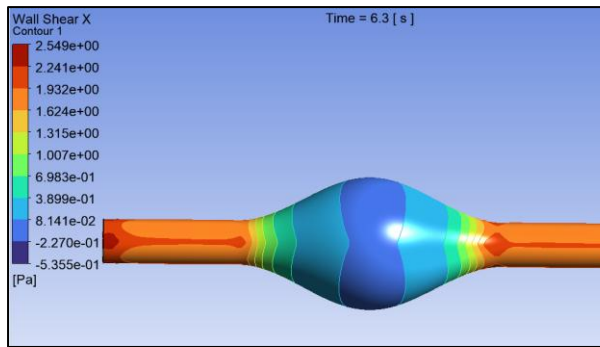


[Figure 2.5]: *Velocity vector plot at low flow condition.*

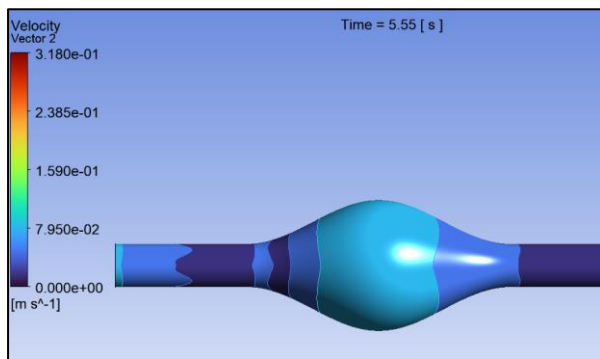


[Figure 2.6]: *Velocity vector plot at peak systolic flow.*

Transient wall shear stress contours demonstrated substantial temporal variation throughout the cardiac cycle. Elevated wall shear stress values occurred during peak flow conditions, particularly near the inlet and outlet regions, while low and occasionally reversed wall shear stress values developed within the aneurysm bulge during decelerating flow conditions. These results highlighted the complex haemodynamic environment generated by pulsatile flow within aneurysmal geometries.



[Figure 2.7]: Wall shear stress contour during peak flow.



[Figure 2.8]: Wall shear stress contour during lowest flow.

The transient CFD-Post animations further illustrated this changing nature of the flow field, particularly the periodic formation, growth and downstream transport of the vortical recirculation structures which form and dissipate within the aneurysm.

Discussion & Conclusions

The CFD simulations highlighted the importance that inlet boundary conditions and transient flow behaviour have on the haemodynamics within an Abdominal Aortic Aneurysm. Across all simulations, the aneurysm geometry produced a characteristic high-velocity central jet surrounded by regions of low-velocity recirculating flow within the aneurysm bulge. These recirculation regions were consistently identified through the velocity vector plots though became increasingly pronounced under more physiologically realistic inlet conditions.

The steady uniform inlet simulation provided a useful baseline approximation of the flow behaviour within the aneurysm. Although the general flow structures and wall shear stress distributions were captured successfully, the assumption of a constant uniform inlet velocity simplified the inlet development of the flow and therefore did not fully represent physiological blood flow conditions. The introduction of the parabolic inlet profile in Part 3.2 produced a more realistic developed inlet flow field, resulting in a more concentrated centreline jet and stronger near-wall velocity gradients at the inlet. However, while local flow behaviour changed slightly, the overall haemodynamic patterns within the aneurysm remained largely similar to the uniform inlet simulation.

The transient periodic simulation however was the most physically realistic approximation of the flow. The pulsatile inlet waveform had variation in the velocity field, wall shear stress distributions and vortex structures throughout the cardiac cycle. The CFD-Post animations clearly demonstrated the formation, persistence and returning downstream transport of the vortical recirculation structures within the aneurysm bulge during different phases of the cycle.

Wall shear stress was strongly dependent on both vessel geometry and flow behaviour. Regions of elevated wall shear stress occurred primarily near the inlet and outlet where velocity gradients were largest, while significantly lower wall shear stress values developed within the aneurysm bulge. The x -component wall shear stress contours additionally identified regions of negative shear stress associated with reversed near-wall flow caused by recirculation.

Overall, there is clear significance to these disturbed haemodynamic conditions

observed, as abnormal wall shear stress environments are associated with degradation of the arterial wall and increased rupture risk in abdominal aortic aneurysms. Previous studies have shown that low and oscillatory wall shear stress can promote endothelial dysfunction, thrombus formation and weakening of the vessel wall, particularly in regions where recirculating flow increases blood residence time near the arterial surface [4].

The transient simulations further demonstrated how pulsatile inlet conditions generated time-dependent vortical structures and continuously varying wall shear stress patterns throughout the cardiac cycle. In comparison with the steady-state simulations, the transient analysis provided a more realistic representation of physiological blood flow behaviour within the aneurysm. Mutlu et al. state that transient CFD analysis is necessary for accurately modelling pulsatile cardiovascular flow and for evaluating haemodynamic quantities such as wall shear stress over time [4].

Bibliography

[1]

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[4]

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